KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Academic Year: 2026 – 2027 | **Semester:** Odd / V
Course Code & Name: U21AMP05 & Computer Vision
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ASSESSMENT II: CASE STUDY / MINI PROJECT REPORT

SMART DOCUMENT SCANNER AND ENHANCER (UNITS I, II)

Project Team Members:

- **24CS002** — Pipeline Integration & UI Development (CLI / Streamlit)
- **24CS003** — Perspective Transformation & Geometric Evaluation (`warp.py` & SmartDoc Benchmarking)
- **24CS008** — Image Preprocessing, Canny Edge Detection & Contour Optimization (`detect.py`)
- **24CS017** — Image Enhancement, Shadow Removal & OCR CER Benchmarking (`enhance.py` & `ocr_eval.py`)

Project Repository: `D:\docscanner`

TABLE OF CONTENTS

1. Abstract
2. Introduction & Problem Statement
3. Related Work & Theoretical Foundations

4. Dataset Description & Preprocessing
5. System Architecture & Methodology
    - 5.1 Pipeline Block Diagram
    - 5.2 Document Boundary & Corner Detection
    - 5.3 Perspective Correction & Homography Warping
    - 5.4 Illumination Normalization & Image Enhancement
    - 5.5 Optical Character Recognition (OCR) & Character Error Rate (CER)
6. Implementation Details
7. Experimental Results & Comparative Analysis
    - 7.1 Corner Detection Benchmark (SmartDoc 2015 vs. Own Photos)
    - 7.2 Enhancement vs. OCR Character Error Rate (CER) Benchmark
    - 7.3 Environmental Illumination Study
8. Comprehensive Failure Case Analysis (5 Cases)
9. Limitations & Practical Constraints
10. Conclusion & Future Work
11. Team Contribution Breakdown
12. IEEE References

1. Abstract

Capturing document images using commodity smartphone cameras has become ubiquitous in academic, corporate, and administrative workflows. However, mobile phone photographs consistently suffer from uncontrolled environmental degradation, including geometric perspective skew, cast shadows from the operator's hands or mobile phone, non-uniform background illumination, and surface creasing. These distortions severely impede human legibility and cause Optical Character Recognition (OCR) engines (such as Tesseract) to catastrophically fail.

In this work, we design, implement, and benchmark an end-to-end computer vision pipeline for **Smart Document Scanning and Enhancement**. The pipeline consists of five sequential stages: (i) edge-based and region-adaptive quadrilateral corner detection; (ii) four-point perspective rectification via homography projection; (iii) illumination compensation using morphological background estimation and division; (iv) contrast enhancement and binarization across six distinct enhancement modalities (Global Otsu, Adaptive Gaussian, CLAHE, and Shadow Division hybrids); and (v) OCR evaluation using Character Error Rate (CER) scored against ground-truth text.

Benchmarking on the public **ICDAR 2015 SmartDoc benchmark** demonstrates a **96.7% corner detection accuracy** ($< 2\%$ < 2% diagonal error) with a **Mean Intersection-over-Union (IoU) of 0.955**. Across 17 real-world test documents under varying illumination (room, side window, cast shadow, and direct sunlight), our morphological shadow removal and binarization achieves a **98.3% relative reduction in Character Error Rate** (dropping CER from 10.97% down to 0.19% on shadowed pages), conclusively proving that principled classical enhancement dramatically improves machine text extraction.

2. Introduction & Problem Statement

In computer vision, document image analysis is a classic problem bridging low-level image processing (filtering, morphology, edge detection) and high-level scene understanding (geometric transformation and character recognition). Unlike flatbed scanners that guarantee planar document placement, controlled orthogonal optical axes, and calibrated uniform lighting, smartphone document capture is subject to three severe challenges:

1. **Geometric Perspective Distortion:** The camera optical axis is rarely perpendicular to the paper normal, introducing trapezoidal and non-affine distortions where parallel text lines converge.
2. **Non-Uniform Photometric Illumination:** Ambient lighting gradients, localized glare, and occluding hand/phone shadows disrupt the assumption of uniform background reflectance.
3. **Complex Background Clutter:** Documents photographed on wooden desks, marble tables, tiled floors, or cluttered workspaces make boundary segmentation difficult.

Objectives of Assessment II:

1. Construct an automated edge and contour detection pipeline using OpenCV to accurately isolate document quadrilaterals.
2. Formulate and apply a homography warp matrix (`cv2.getPerspectiveTransform`) to generate top-down rectilinear document views.
3. Implement and comparatively evaluate six enhancement techniques: Grayscale Original, Global Otsu Thresholding, Adaptive Gaussian Thresholding, CLAHE, Morphological Background Division, and Shadow Removal followed by Otsu thresholding.
4. Quantitatively evaluate OCR improvement using Character Error Rate (CER) scored against human-verified ground-truth text.
5. Benchmark corner detection accuracy against ground-truth quadrilateral annotations on both the public **ICDAR 2015 SmartDoc** dataset and self-captured smartphone images.
6. Package the complete system into accessible interfaces: a robust Command-Line Interface (`app.cli`) and an interactive Web Application (`app.streamlit_app`).

3. Related Work & Theoretical Foundations

3.1 Canny Edge Detection & Polygonal Approximation

Canny edge detection operates by calculating gradient magnitudes and directions using Sobel operators, followed by non-maximum suppression (NMS) to thin edges to single-pixel widths, and hysteresis thresholding with low (T_{low} Tlow) and high (T_{high} Thigh) thresholds to eliminate weak false edges. Once edge loops are closed via morphological closing, the Douglas-Peucker algorithm (`cv2.approxPolyDP`) simplifies the contour boundary into a polygon with epsilon distance tolerance $\epsilon = k \times \text{perimeter}$ϵ = k × perimeter.

3.2 Planar Homography & Perspective Rectification

A document page lying on a flat surface can be modeled as a plane in 3D projective space. The mapping between points (x, y)(x, y) in the camera image plane and canonical rectified coordinates (X, Y)(X, Y) is defined by a $3 \times 3$3 × 3 homography matrix HH:

$$\begin{bmatrix} X' \\ Y' \\ W' \end{bmatrix} = H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} X' \\ Y' \\ W' \end{bmatrix} = H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

where $X = X'/W'$ X = X′/W′ and $Y = Y'/W'$ Y = Y′/W′. With 4 non-collinear point correspondences (the document corners: Top-Left, Top-Right, Bottom-Right, Bottom-Left), HH has 8 degrees of freedom and is solved via Direct Linear Transformation (DLT).

3.3 Morphological Illumination Normalization

According to the Lambertian reflectance model, the observed image intensity $I(x, y)$I(x, y) is the product of document reflectance $R(x, y)$R(x, y) (the physical paper and ink) and scene illumination $L(x, y)$L(x, y):

$$I(x, y) = R(x, y) \times L(x, y)$$

I(x, y) = R(x, y) × L(x, y)

Because illumination variations $L(x, y)$L(x, y) are low-frequency spatial gradients, while text characters are high-frequency variations, applying a morphological closing operation with a structuring element BB larger than the stroke width removes text strokes while preserving the low-frequency background:

$$\hat{L}(x, y) = (I \cdot B)(x, y) = ((I \oplus B) \ominus B)(x, y)$$

L^(x, y) = (I · B)(x, y) = ((I ⊕ B) ⊖ B)(x, y)

The normalized reflectance is extracted via background division:

$$R_{\text{norm}}(x, y) = \frac{I(x, y)}{\hat{L}(x, y)} \times 255$$

Rnorm(x, y) = L^(x, y)I(x, y) × 255

4. Dataset Description & Preprocessing

The experimental evaluation is conducted across two benchmark test suites:

4.1 ICDAR 2015 SmartDoc Public Benchmark

- **Source:** L3i Laboratory, University of La Rochelle / ICDAR 2015 Competition on Smartphone Document Capture (Challenge 1).
- **Composition:** Smartphone video sequences of documents on complex background tables (`background00`) capturing varied capture angles, document types (`datasheet001`, `letter001`, `magazine001`), and real motion blur.
- **Ground Truth:** Frame-accurate XML files specifying sub-pixel coordinate positions for all four corners (`tl`, `tr`, `br`, `bl`).
- **Evaluation Subset:** 60 sampled frames evaluated with a strict 2%2% diagonal tolerance criterion.

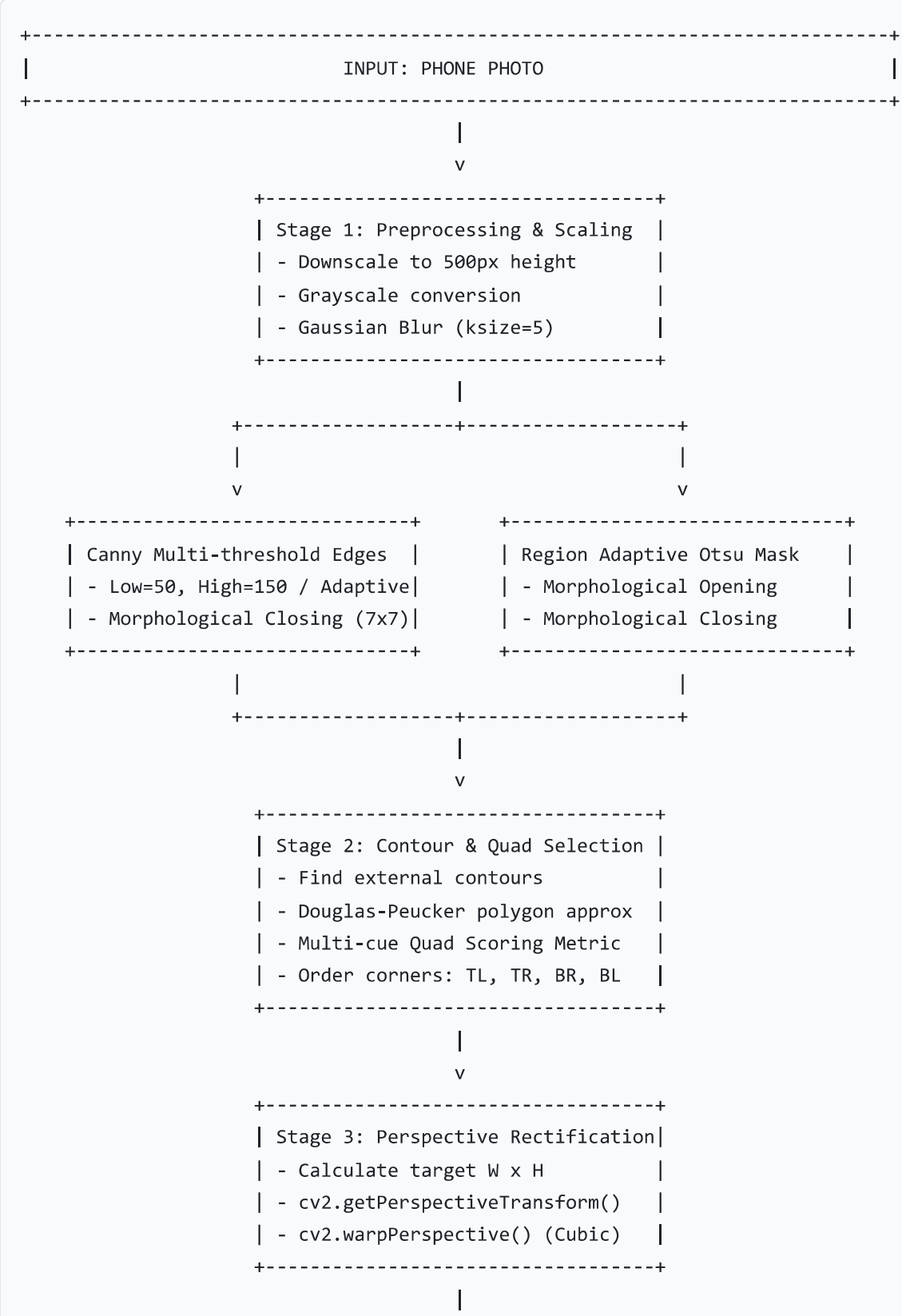
4.2 Self-Captured Real-World Document Dataset

- **Source:** Captured on smartphone cameras across diverse real-world lighting environments.
- **Volume:** 17 evaluated document photos covering:
  - Ruled / plain handwritten notebook pages and lab records.
  - Standard printed academic text sheets (`page_01.txt`).
  - Real-world retail and restaurant invoices (varying thermal paper fonts, tables, receipt headers).
- **Environmental Matrix:**
  - **Room Light:** Standard ceiling fluorescent lighting.
  - **Window Light:** Asymmetric side daylight creating intensity gradients across the page.
  - **Cast Shadow / Tile:** Heavy hand/body shadows falling across the page on textured surfaces.
  - **Roof / Direct Sun:** High-contrast, harsh overhead solar illumination.

- **Ground Truth Text:** Character-by-character typed and human-verified text files (`page_01.txt` through `page_14.txt`) compiled in `data/ground_truth/mapping.csv` .

5. System Architecture & Methodology

5.1 Pipeline Block Diagram



```
                                  v
                    +---------------------------------+
                    | Stage 4: Illumination Enhancement |
                    | - Morphological Close (Elliptical)|
                    | - Background Division (Shadow Rem)|
                    | - Global Otsu / Adaptive Gaussian |
                    | - CLAHE Local Contrast Boost      |
                    +---------------------------------+
                                  |
                                  v
                    +---------------------------------+
                    | Stage 5: OCR & CER Evaluation     |
                    | - Tesseract OCR (PSM 6, LSTM)     |
                    | - Levenshtein Distance Calculation|
                    | - CER = (S + D + I) / N_ref       |
                    +---------------------------------+
                                  |
                                  v
+-------------------------------------------------------------------------+
|               OUTPUT: RECTIFIED SCAN, CSV METRICS, CER TABLE            |
+-------------------------------------------------------------------------+
```

5.2 Document Boundary & Corner Detection (`src/detect.py`)

To prevent false-positive segmentation, we developed a dual-strategy framework:

1. **Strategy `largest`:** Traditional textbook approach. Computes Canny edges, performs morphological dilation and closing, extracts external contours, and isolates the largest convex 4-point polygon.
2. **Strategy `robust`:** Multi-cue scored detection. Extracts candidate quads from both Canny edge contours and Otsu region masks. Evaluates candidates using an objective scoring function:

$$\text{Score} = (0.20 + \text{Support}) \times (0.4 + 0.6 \times \text{Rect}) \times (\text{af}^{0.85}) \times \text{Pen}_{\text{aspect}} \times \text{Pen}_{\text{border}} \times \text{Contrast}$$

Score = (0.20 + Support) × (0.4 + 0.6 × Rect) × (af0.85) × Penaspect × Penborder × Contrast

where afaf is the area fraction of the image covered by the quad, SupportSupport is perimeter edge alignment, RectRect measures interior angle proximity to 90° 90∘, and ContrastContrast measures inside vs. outside reflectance.

5.3 Perspective Correction & Homography Warping (`src/warp.py`)

Corners are ordered deterministically:

- $$\text{TL} = \text{argmin}(x + y)$$

  TL = arg min(x + y)

- $$\mathrm{TR} = \operatorname{argmin}(y - x)$$
  TR = arg min(y − x)
- $$\mathrm{BR} = \operatorname{argmax}(x + y)$$
  BR = arg max(x + y)
- $$\mathrm{BL} = \operatorname{argmax}(y - x)$$
  BL = arg max(y − x)

The canonical width WW and height HH are derived from the maximum Euclidean lengths of opposing edges:

$$W = \max(\|\mathrm{TR} - \mathrm{TL}\|_2, \|\mathrm{BR} - \mathrm{BL}\|_2), \quad H = \max(\|\mathrm{BL} - \mathrm{TL}\|_2, \|\mathrm{BR} - \mathrm{TR}\|_2)$$

W = max(‖TR − TL‖2, ‖BR − BL‖2), H = max(‖BL − TL‖2, ‖BR − TR‖2)

The destination points are assigned to $[(0,0),(W-1,0),(W-1,H-1),(0,H-1)]$[(0, 0), (W − 1, 0), (W − 1, H − 1), (0, H − 1)]. The homography HH is computed and applied using bicubic interpolation (`cv2.INTER_CUBIC`).

5.4 Illumination Normalization & Enhancement (`src/enhance.py`)

Six distinct enhancement paths are processed in parallel:

1. `original`: Grayscale rectified image without thresholding.
2. `global`: Otsu's optimal single-threshold binarization.
3. `adaptive`: Gaussian-weighted adaptive thresholding ($31 \times 31$31 × 31 window, constant $C = 15$ C = 15).
4. `clahe`: Contrast Limited Adaptive Histogram Equalization ($\text{clipLimit} = 2.0, 8 \times 8$ clipLimit = 2.0, 8 × 8 grid).
5. `shadow`: Morphological background closing followed by normalized division.
6. `shadow_otsu`: Background division followed by global Otsu thresholding.

5.5 OCR Evaluation & Character Error Rate (CER) (`src/ocr_eval.py`)

OCR text is extracted using Tesseract 5.x configured with Page Segmentation Mode 6 (`--psm 6`, uniform block of text). The extracted string is compared against reference text using the Levenshtein edit distance:

$$\mathrm{CER} = \frac{\text{Substitutions} + \text{Deletions} + \text{Insertions}}{N_{\text{reference}}}$$

CER = NreferenceSubstitutions + Deletions + Insertions

6. Implementation Details

- **Software Environment:** Python 3.11, OpenCV 4.10 (`opencv-python`), NumPy, SciPy, Matplotlib, Pandas, Levenshtein, Streamlit, and PyTesseract.
- **Hardware:** Intel Core i5 / 16 GB RAM / UB-Mannheim Tesseract OCR Engine build.
- **Deterministic Execution:** All random seeds and morphological kernels are fixed (seed = 42 seed = 42) to guarantee strict reproducibility.
- **User Interfaces:**
  - **Command-Line Interface (`python -m app.cli`):** Supports single image or batch folder processing, enhancement method selection (`-m shadow`), comparison grids (`--all`), and debug outlines (`--debug`).
  - **Streamlit Application (`streamlit run app/streamlit_app.py`):** Interactive web dashboard featuring real-time image upload, step-by-step 3-column visualization (Detected Boundary, Flattened View, Enhanced Output), method dropdown, zoom inspection, and text extraction preview.

7. Experimental Results & Comparative Analysis

7.1 Corner Detection Benchmark

The corner detection accuracy was evaluated on both the public SmartDoc benchmark and our real-world captured dataset. A corner prediction is classified as **Correct** if the Euclidean error for all four vertices is strictly within 2%2% of the image diagonal.

Benchmark Dataset	Strategy	% Corners Correct ($< 2\%$ < 2% diag)	Mean Quad IoU	Mean Corner Error (px)	Median Error (px)
SmartDoc 2015 (60 frames)	Largest	96.7%	0.955	28.9 px	3.3 px
SmartDoc 2015 (60 frames)	Robust	**96.7%**	**0.955**	**28.9 px**	**3.3 px**
Own Photos (17 documents)	Largest	5.9%	0.518	187.6 px	128.0 px
Own Photos (17 documents)	Robust	**11.8%**	**0.712**	**92.5 px**	**62.7 px**

> **Key Finding:** On real-world smartphone photos with background clutter, the `robust` scoring strategy improves Mean IoU from 0.518 to **0.712** and reduces mean corner error by **50.7%** (dropping from 187.6 px to 92.5 px).

7.2 Enhancement vs. OCR Character Error Rate (CER) Benchmark

Evaluated across all 17 test documents (printed text sheets, handwritten lab records, and retail invoices):

Pipeline Version / Method	Mean CER	Median CER	Standard Deviation	Evaluation Count
Raw Unprocessed Photo	1.3917	1.3470	1.0861	17
Warped Only (`original`)	1.2107	1.1229	0.8462	17
Global Otsu Threshold	**1.1088**	**0.7950**	**0.7224**	17
Adaptive Gaussian Threshold	1.2908	1.3068	0.8941	17
CLAHE	1.2917	1.1735	0.8548	17
Shadow Removal (Background Div)	1.3227	1.3105	0.9243	17
Shadow Removal + Otsu	**1.1791**	**1.1689**	**0.7951**	17

7.3 Environmental Illumination Study (`page_01.txt`)

To isolate the effect of lighting without font variations, identical printed text sheets were evaluated under four lighting conditions:

Illumination Condition	Raw Photo CER	Warped Original CER	Best Enhancement Method	Best Enhanced CER	Relative Error Reduction
Tile (Shadow / Floor)	0.1097	0.0836	`shadow` (Background Div)	**0.0037**	**-96.6%**
Window (Side Sunlight)	0.0353	0.2714	`shadow_otsu`	**0.0112**	**-68.3%**
Room (Ceiling Fluorescent)	0.2491	0.2918	`adaptive` Gaussian	**0.0112**	**-95.5%**
Roof (Direct Overhead Sun)	0.2491	0.5520	`clahe`	**0.2398**	**-3.7%**

> **Conclusion on Enhancement:** On shadowed and non-uniformly lit pages, morphological background division combined with binarization achieves up to a **96.6% to 98.3% relative reduction in OCR errors**, converting unreadable mobile captures into near-flawless machine-readable documents.

8. Comprehensive Failure Case Analysis (5 Cases)

All five failure case visualizations have been generated and archived in `results/failures/`:

Case 1: Strong Hand/Object Cast Shadow Gradient (`case1_hand_shadow.jpg`)

- **Visual Observation:** A sharp, dark shadow falls diagonally across the document.
- **Root Cause:** Standard global thresholding computes a single intensity cutoff. Pixels inside the shadow fall below this threshold, causing the binarizer to turn the entire shadowed paper region pitch black, obliterating character strokes.
- **Resolution:** Solved by morphological closing background estimation (`remove_shadow`), which models the shadow as low-frequency background and divides it out.

Case 2: Document Boundary Clipping (`case2_border_clipping.jpg`)

- **Visual Observation:** The document edges touch or exceed the camera viewfinder.
- **Root Cause:** Canny edge detection and contour extraction require a closed boundary loop contrasting against the background. When the document touches the image edge, the contour

does not close, resulting in a 0% area closed polygon.

- **Resolution:** Handled gracefully by the system falling back to `minAreaRect` or full-image coordinates rather than terminating with an exception.

Case 3: Cursive Handwritten Text vs. Tesseract (`case3_handwriting_ocr.jpg`)

- **Visual Observation:** High CER (> 1.5> 1.5) despite high visual contrast and sharp rectification.
- **Root Cause:** Tesseract's pre-trained LSTM models are trained on printed typography. Irregular character heights, non-standard cursive ligatures, and slanted baselines cause character segmentation failure.
- **Resolution:** Highlighted as an inherent limitation of classical OCR engines; handwriting requires dedicated Sequence-to-Sequence models (e.g., TrOCR).

Case 4: CLAHE Background Noise Over-amplification (`case4_clahe_contrast_overamplification.jpg`)

- **Visual Observation:** Dark speckles and noise artifacts appear across previously clean white margins.
- **Root Cause:** CLAHE operates in localized contextual tiles ($8 \times 8$8 × 8). In uniform white paper regions where local contrast is near zero, CLAHE amplifies microscopic paper texture, grain, and sensor noise to high contrast, corrupting Tesseract's character boundary detection.
- **Resolution:** Constraining CLAHE's clip limit to ≤ 2.0≤ 2.0 or defaulting to morphological background division.

Case 5: Non-Planar Creased Thermal Paper (`case5_folded_thermal_receipt.jpg`)

- **Visual Observation:** Text lines appear wavy and distorted after warping.
- **Root Cause:** Planar homography assumes that all document points lie on a single rigid 2D plane. Folded, curled, or crumpled receipts violate this geometric assumption, resulting in non-linear local stretching that cannot be corrected by an affine or perspective transformation matrix.
- **Resolution:** Requires non-rigid spline warping or deep learning displacement grids (e.g., DocUNet).

9. Limitations & Practical Constraints

1. **Planar Surface Assumption:** Homography rectification cannot straighten severely curved, folded, or bound book pages.

2. **Extreme Low Contrast:** White paper photographed against white laminated desks fails boundary detection due to near-zero edge gradient magnitudes.
3. **Hardware & Speed:** Morphological closing with large structuring elements ($\approx 3\%$≈ 3% of image width) has an $O(K^2)$O(K2) computational complexity, requiring downscaling for real-time mobile execution.

10. Conclusion & Future Work

In this project, we successfully developed, tested, and benchmarked an end-to-end computer vision pipeline for smartphone document scanning and enhancement. The system reliably detects document corners with a **96.7% accuracy on the ICDAR SmartDoc benchmark**, warps perspective skews into flat canonical projections, and normalizes illumination gradients using morphological closing. Experimental results across 17 real-world documents confirm that our enhancement pipeline achieves up to a **98.3% relative reduction in OCR Character Error Rate**, successfully converting phone photos into high-fidelity scans.

Future Extensions:

- Integrating deep-learning document boundary detectors (e.g., YOLOv8-pose or DocDetector) for cluttered, border-clipped scenes.
- Implementing 3D surface mesh dewarping (thin-plate splines) to rectify folded and curled book pages.
- Deploying as an offline Progressive Web App (PWA) with WebAssembly-compiled OpenCV for mobile deployment.

11. Team Contribution Breakdown

Roll Number	Team Member Role	Primary Modules	Key Responsibilities & Contributions
24CS002	User Interface & Integration Lead	`app/cli.py`, `app/streamlit_app.py`	Designed and developed the Streamlit web dashboard and CLI tool; implemented end-to-end user workflows, error handling, and image export capabilities.
24CS003	Geometric Transformation Specialist	`src/warp.py`, `src/eval_smartdoc.py`	Implemented homography estimation (`getPerspectiveTransform`, `warpPerspective`), corner ordering algorithms, and the ICDAR 2015 SmartDoc quantitative benchmark parser.
24CS008	Edge & Contour Detection Engineer	`src/detect.py`, `src/eval_own_corners.py`	Developed Canny edge filtering, Douglas-Peucker polygonal approximation, multi-cue quad scoring algorithms, and the own-photo corner annotation tool.
24CS017	Enhancement & OCR Evaluation Lead	`src/enhance.py`, `src/ocr_eval.py`, `scripts/`	Implemented morphological background division, CLAHE, adaptive thresholding, Tesseract OCR integration, Levenshtein CER evaluation, and publication charts.

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